

Delayed Anaphylaxis to Red Meat in Patients with IgE Specific for Galactose alpha-1,3-Galactose (alpha-gal)

Anaphylaxis is a severe allergic reaction that can be rapidly progressing and fatal. In instances where the triggering allergen is not known, establishing the etiology of anaphylaxis is pivotal to long-term risk management. Our recent work has identified a novel IgE antibody (Ab) response to a mammalian oligosaccharide epitope, galactose-alpha-1,3-galactose (alpha-gal), that has been associated with two distinct forms of anaphylaxis: (1) immediate onset anaphylaxis during first exposure to intravenous cetuximab, and (2) delayed onset anaphylaxis 3–6 h after ingestion of mammalian food products (e.g., beef and pork). The results of our studies strongly suggest that tick bites are a cause, if not the only significant cause, of IgE Ab responses to alpha-gal in the southern, eastern and central United States. Patients with IgE Ab to alpha-gal continue to emerge and, increasingly, these cases involve children. This IgE Ab response cross-reacts with cat and dog but does not appear to pose a risk for asthma; however, it may impair diagnostic testing in some situations.